

CS471 - Web Technologies (Laboratory)

Lab 4: HTML and CSS (Part 1)

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GitHub Repository: github.com/lujain157/CS471-Labs

Overview

This lab covers fundamental HTML tags, styling with CSS (inline, internal, and external), W3C HTML5 validation, and integrating HTML/CSS into Django using static and URL template tags and template inheritance.

Task 1 — Fundamental HTML Document and Tags



Figure 1: index.html rendered with basic HTML tags (headings, paragraph, link, line breaks)

The index.html file was built with fundamental HTML tags including headings, paragraphs, a hyperlink, a navigation menu, and line breaks, and opened directly in the browser.

Task 2 — Styling with CSS (Inline, Internal, External)

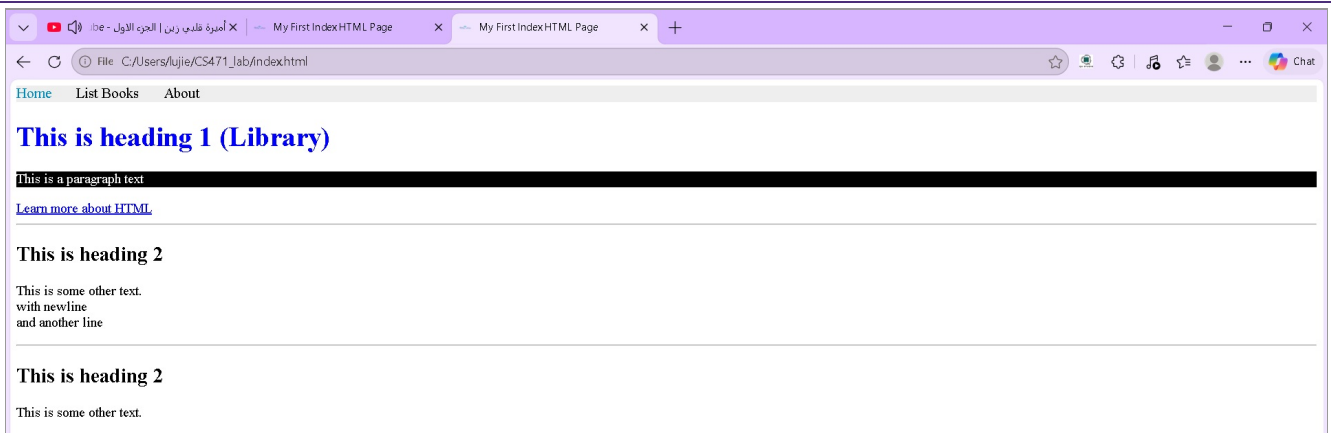


Figure 2: index.html after applying inline CSS, internal CSS, and the external styles.css file

Inline CSS was applied to the heading and paragraph, internal CSS was added in list_books.html, and an external styles.css file was linked to style the navigation bar across all pages.

Task 3 — W3C HTML5 Validation

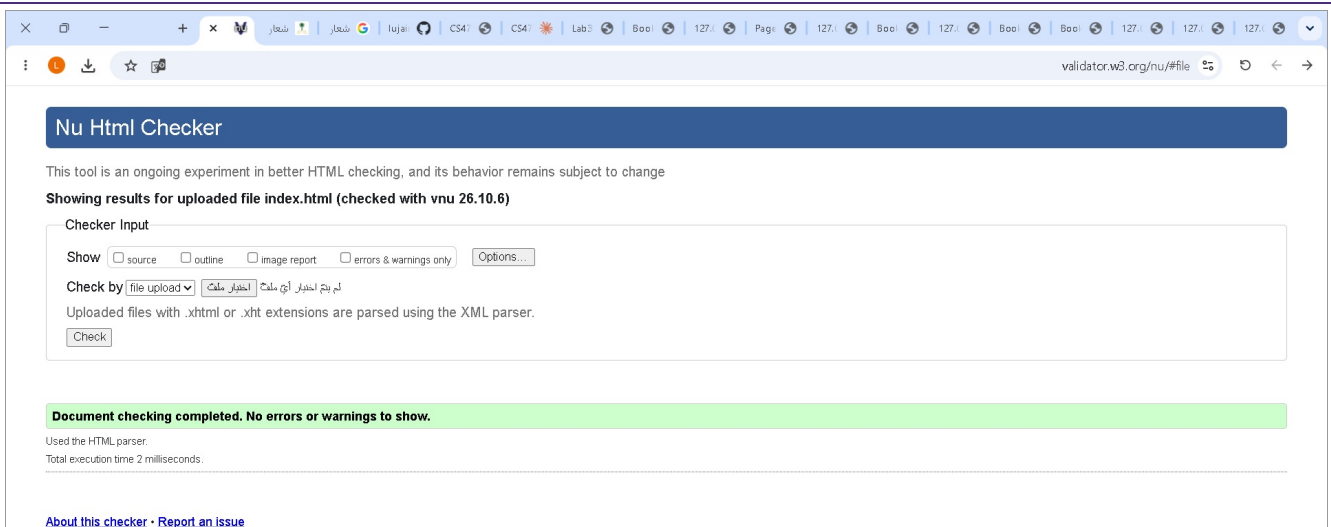


Figure 3: W3C Nu Html Checker result for index.html — no errors or warnings

The index.html file was uploaded to the W3C HTML5 validation service, and the check confirmed no errors or warnings, verifying correct HTML5 syntax.

Task 4 — HTML and Static Files in Django

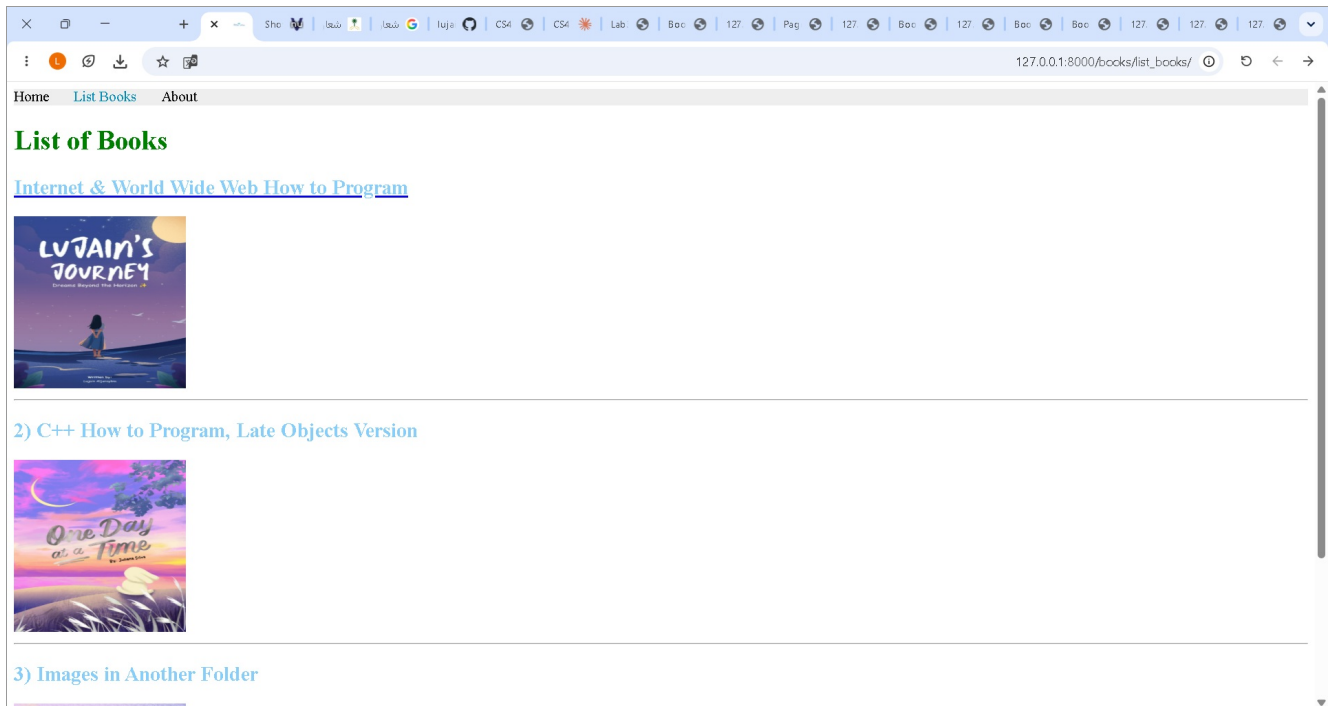


Figure 4: List of Books page served by Django at /books/list_books/, with static CSS and images loaded

The HTML files and static files (CSS and images) were moved into the Django apps folder, static settings were configured, and views/URLs were added so the pages render correctly through Django with static and URL template tags.

Task 5 — Template Inheritance

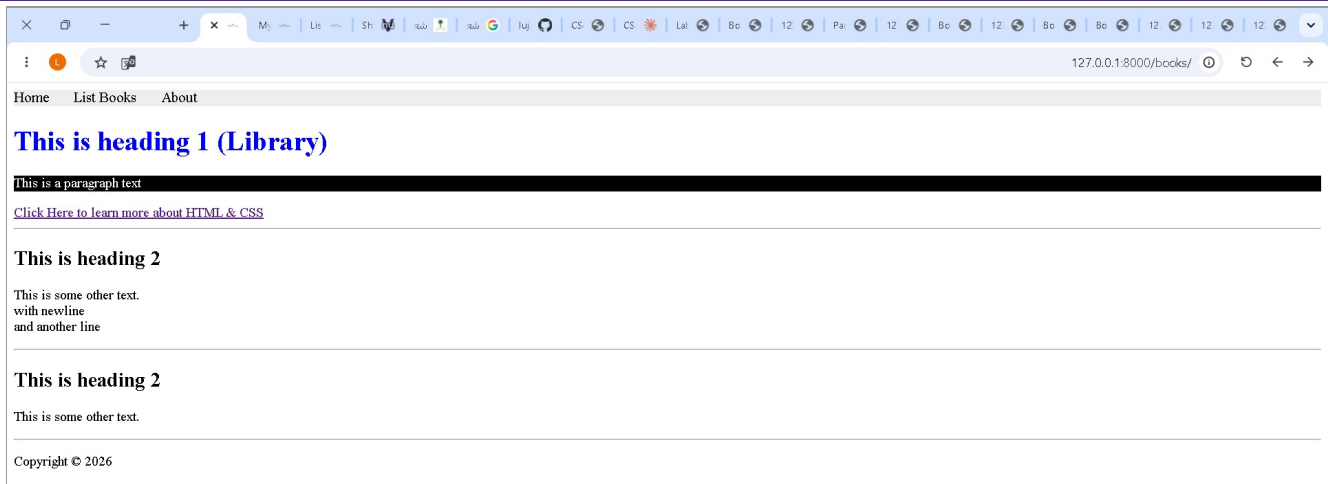


Figure 5: index page at /books/ rendered using base.html layout with header and footer includes

A base layout (base.html) with header and footer includes was created using Django block and extends tags, and index.html was updated to extend this layout, avoiding repetition of the common page structure.

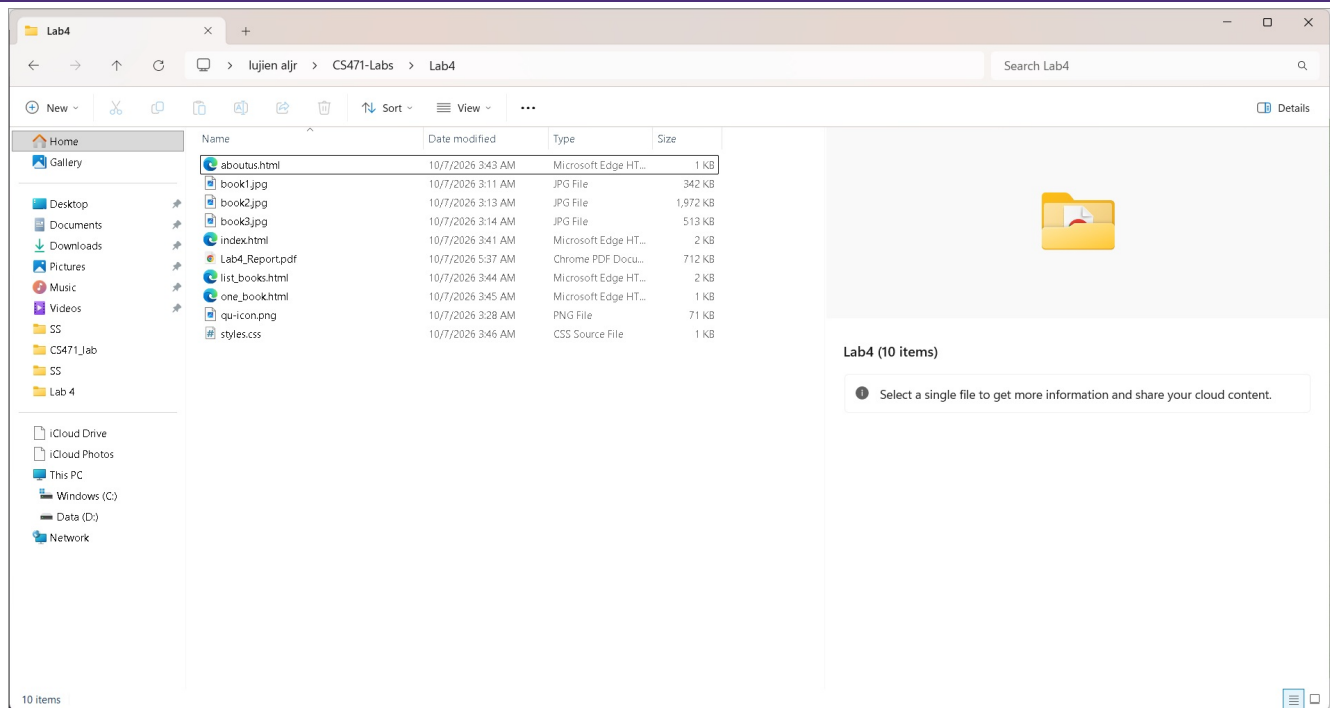


Figure 6: Lab 4 project folder contents in File Explorer (CS471-Labs/Lab4)

Steps Performed

- Created index.html, aboutus.html, list_books.html, and one_book.html with fundamental HTML tags.
- Applied CSS using inline, internal, and external (styles.css) methods.
- Validated index.html using the W3C HTML5 validation service with no errors.
- Configured Django static files settings and moved CSS/images into the apps/static folder.
- Added Django views and URL patterns, and used static and url template tags in the HTML files.
- Built a base layout with header/footer includes and used template inheritance across pages.

Conclusion

The lab successfully demonstrated building HTML pages, styling them with CSS, validating HTML5 syntax, and integrating static files and template inheritance into a Django project.